

Supplement 2: QUADAS-3 Tool

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QUADAS-3

QUADAS-3 aims to help in the evaluation of the risk of bias and concerns regarding applicability for each set of estimates of test accuracy included in your review. The tool includes six phases. Phases 1 and 2 are completed at the review level, phases 3 and 4 at the study level, and phase 5 and 6 are completed for each selected set of estimates. We recommend including the review level phases as part of your review protocol. Table 1 summarises the phases included in the QUADAS-3 tool with details of when each phase should be completed and which section of the tool should be completed for each phase.

Please **read the QUADAS-3 Explanation & Elaboration** report before using QUADAS-3. This provides a detailed overview of how to complete each phase. Consider whether any review-specific guidance for individual signalling questions is needed.

Table 1: Overview of the phases included in the QUADAS-3 tool with details of when each phase should be completed and which section of the tool should be completed for each phase

Phase	When to complete	Section to complete
Phase 1: State the systematic review synthesis question(s)	Once per review	Table 2
Phase 2: Define the ideal test accuracy trial for each synthesis question	Once per review	Table 4
Phase 3: Draw a flow diagram	Once per study	Box 1
Phase 4: Identify the accuracy estimates to assess for risk of bias and applicability	Once per study	Table 5
Phase 5: Assess risk of bias and concerns regarding applicability	For each selected estimate	Individual domain
Phase 6: Overall judgement	For each selected estimate	Summary tables

Phase 1: State the systematic review synthesis question(s) [once per review]

A single review may address more than one synthesis question. The first phase of the QUADAS-3 assessment is to clearly specify the question(s) that will be addressed by the systematic review synthesis. This should also be pre-specified in your review protocol. Consider including the following components: population, index test(s), and target condition. For reviews involving comparative accuracy, we recommend using the QUADAS-C tool in addition to QUADAS-3.

Table 2: Definition of the systematic review synthesis questions

Question 1	
Question 2 (if applicable)	

Phase 2: Define the ideal test accuracy trial for each synthesis question [once per review]

For each synthesis question defined in Phase 1, define the ideal test accuracy trial that could be conducted in a clinical setting. An ideal test accuracy trial should produce unbiased estimates of test accuracy. An ideal trial is a theoretical construct as such a trial design may not be achievable in practice because of ethical and resource constraints.

Defining the ideal test accuracy trial serves as a benchmark for identifying how real-world deviations observed in a particular study may introduce risk of bias or raise concerns regarding applicability.

The ideal test accuracy trial should be based on the following design template, adapted to your review question, and should ensure that the design both minimises the risk of bias and maximises the applicability of the trial to each synthesis question. A test strategy encompasses one or more tests used together but accuracy is estimated for the strategy rather than for the individual test components. Table 3 provides a framework to help you define the ideal test accuracy trial for each synthesis question, recognising that some components may be replicated across individual synthesis questions in a review (for example a review of more than one index test in the *same* population).

Table 3: Framework to define the ideal test accuracy trial. Use this information to help you complete Table 4

Component	General features
Overall	• Pre-specified protocol. • Sufficient total sample size to expect adequate numbers of participants with and without the target condition to generate sufficiently precise estimates of test accuracy (e.g. sensitivity and specificity).
Objective	• Assess the accuracy of the index test(s) in its proposed role and position in the clinical pathway, including the clinical decision that the test is informing.
Participants	• Prospective enrolment of a single group of consecutive or random group of eligible participants suspected of having the target condition. • Participants are a representative sample of the intended-use population, as defined in the synthesis question. • Details of symptoms and prior test results of included patients reported.
Index test	• Index test should be the appropriate technical version of the index test, conducted using recommended instructions and in the intended role in the clinical pathway. • Performed after recruitment of trial participants. • Interpretation of the index test result should not be affected by external information that would not usually be available to the person interpreting the test in its proposed role in practice e.g. information on the reference standard, disease state, or additional test results. • Test conducted by operators who would usually conduct and interpret the test in practice, in a setting representative of the review question. • Definition of threshold(s) to classify test results as positive or negative pre-specified.
Definition of the target condition	• Reference standard and reference standard threshold should correctly classify all participants as having or not having the target condition. • Protocol for undergoing the reference standard is the same for all study participants, and the reference standard is performed in all participants. • Reference standard performed after recruitment of trial participants. • Reference standard interpretation is independent of the index test; those interpreting the reference standard should not be aware of the index test results, and the index test should not be part of the reference standard where a composite reference standard is used. • The time between the index test and reference standard is sufficiently short to be confident that the disease status has not changed.
Analysis	• All participants included in the analysis. • Unit of analysis appropriate for the synthesis question. • Appropriate methods used to calculate estimates of accuracy (e.g. sensitivity and specificity).

For each synthesis question in Table 2, define the ideal trial in Table 4. Use the information in Table 3 as a guide.

Table 4: Definition of the ideal test accuracy trial for each systematic review synthesis question in Table 2

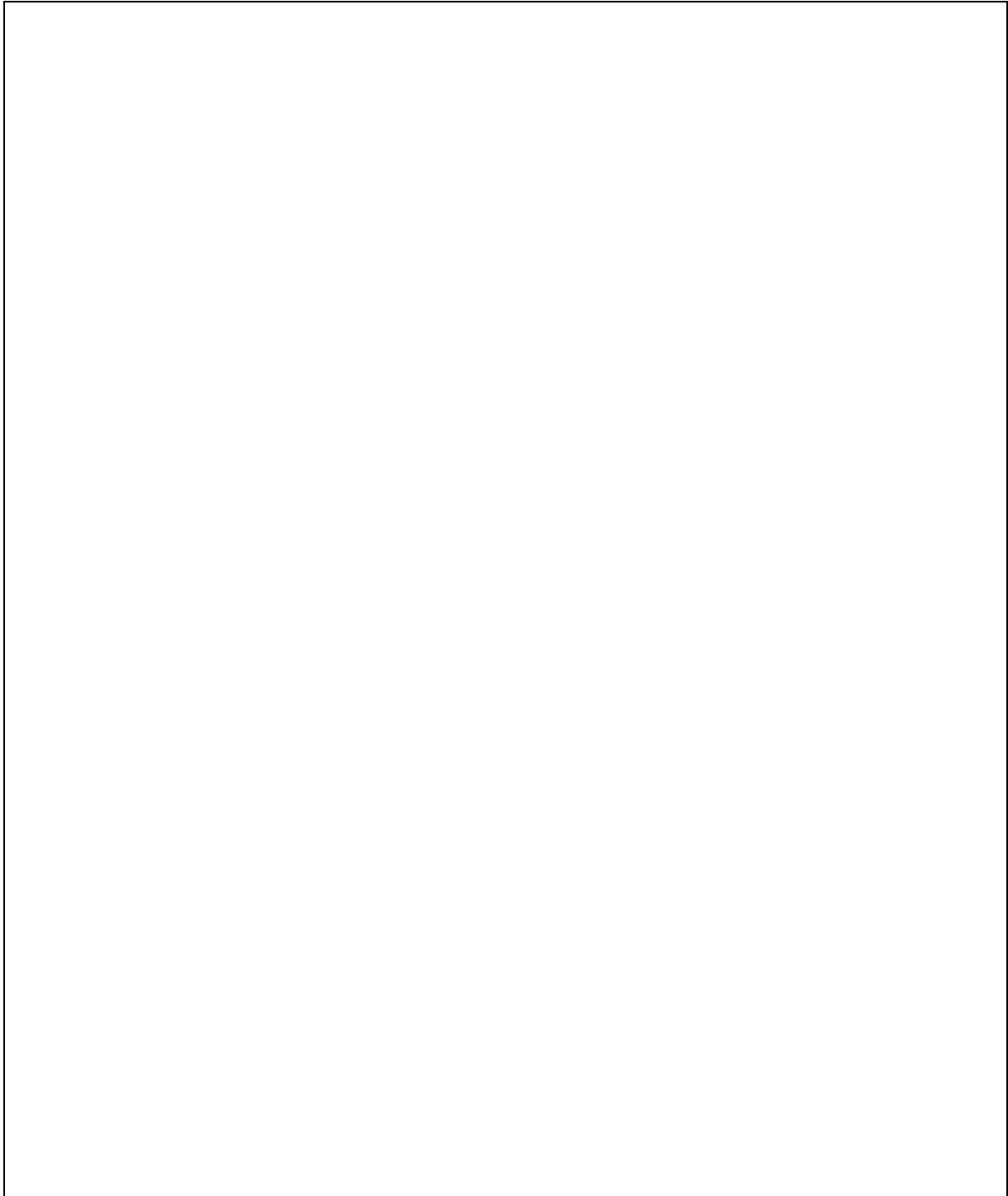
Component	Synthesis Question 1	Synthesis Question 2
Objective		
Participants		
Index Test(s)		
Definition of the target condition		
Analysis		

Phase 3: Draw a flow diagram [once per study]:

Draw a flow diagram for the primary study to provide a visual summary of how participants and test results underlying accuracy estimates progress through a primary study. This would usually be done at the study level and would portray the flow of participants and test results for all estimates specified in Phase 4. You may need to draw separate diagrams for the different synthesis questions.

Reference of paper being assessed:

Box 1: Flow diagram

A large, empty rectangular box with a thin black border, intended for drawing a flow diagram. The box is currently blank.

Phase 4: Identify the numerical accuracy estimates to assess for risk of bias and applicability [once per study]

For each synthesis question, it is often possible to calculate multiple estimates of sensitivity and specificity (i.e. multiple two-by-two tables) from the data presented in a study. Multiple estimates within a specific study may occur due to differences in study subgroups, index tests, definitions of the target condition, reference standards, units of analysis, and thresholds. The potential risk of bias and applicability issues may differ according to the estimate.

Not all estimates reported in a primary study will be relevant for your review; you should only extract and assess those estimates that are relevant to one of your synthesis questions (defined in Phase 1) and that will contribute to the synthesis (whether narrative or statistical). Deciding which estimates to extract should be considered at the review level and done as part of data extraction – here we ask you to specify which estimates will be included in your review. Only perform QUADAS-3 assessment for numerical estimates that will be included in the review. Please complete Table 5 to identify the numerical accuracy result(s) being assessed.

Table 5: Identification of numerical accuracy estimates(s) to be assessed

	Estimate 1	Estimate 2	Estimate 3
Synthesis Question (e.g. Q1, Q2 from table X)			
Numerical test accuracy result e.g. sensitivity=80% and specificity=76%			
Participants			
Index test			
Index test threshold (if applicable)			
Target condition			
Reference standard			
Unit of analysis (e.g. participant, tumour, lesion, sample)			
Analysis (e.g. analysis method, participants included in analysis)			
Which QUADAS-3 domains will be assessed (Participants, Index Test, Target Condition, Analysis)*			

* After the first estimate, only domains where characteristics are different between estimates need to be assessed

Phase 5: Assess risk of bias and concerns regarding applicability [for each selected estimate]

QUADAS-3 is structured as 4 key domains (Participants, Index Test, Target Condition, Analysis) each rated in terms of the risk of bias. Each domain has a set of signalling questions to help judge risk of bias. Signalling questions are answered as:

- yes (Y)
- probably yes (PY)
- probably no (PN)
- no (N)
- no information (NI)

A risk-of-bias judgement for each domain is based on signalling question answers and is answered as:

- low
- high
- insufficient information (II)

If all signalling questions for a domain are answered “yes” or “probably yes” then risk of bias can be judged “low”. If any signalling question is answered “no” or “probably no” this flags the potential for bias. Review authors then need to use their judgement and review-specific assessment guidance to judge whether the potential issue identified is likely to have influenced the estimates of test accuracy – a study can still be at low risk of bias even if one or more signalling questions are answered no. The “insufficient information” category should be used only when insufficient data are reported to permit a judgment.

Three of the four domains (Participants, Index Test, Target Condition) are also assessed for applicability to the synthesis question. Concern regarding applicability to the synthesis question is also judged as:

- low
- high
- insufficient information

If you have specified more than one estimate to be assessed in Table 5, you will need to identify which assessment relates to which estimate.

DOMAIN 1: PARTICIPANTS

A. Risk of bias	
<i>Note: If participants dropped out or were excluded from a study because they did not receive the index test and/or reference standard, then this should be handled in the analysis domain.</i> Describe how participants were enrolled, e.g. single-gate (a group of participants in whom the diagnosis was not yet known. This could include multiple groups from different locations, e.g. different hospitals) or multi-gate (case-control design – participants with known diagnosis): Describe whether enrolment was prospective or retrospective: Describe how participants were selected for enrolment (consecutive or random): Describe any restrictions to enrolment:	
1.1 Was a single-gate design used?	Y/PY/PN/N/NI
1.2 Were participants prospectively enrolled?	Y/PY/PN/N/NI
1.3 Was a consecutive or random sample of participants included?	Y/PY/PN/N/NI
1.4 Is the study group a representative sample of the intended-use population?	Y/PY/PN/N/NI
Risk that the selection of participants has introduced bias	LOW/HIGH/II
Rationale for judgement	
B. Concerns regarding applicability to the systematic review synthesis question	
Concern that the included participants do not match those in the ideal test accuracy trial	LOW/HIGH/II
Rationale for judgement Describe how differences between the included participants (e.g. presentation, prior testing, setting, intended use of index test) and the ideal test accuracy trial defined in Table 4 have led to this judgement	

Y = yes; PY = probably yes; N = no; PN = probably no; NI = no information; II = insufficient information

DOMAIN 2: INDEX TEST

A. Risk of bias	
<i>Describe the index test:</i> <i>Describe whether the index test was conducted and interpreted according to recommended instructions (e.g. following manufacturer's instructions, following protocol, standard clinical methods):</i> <i>Describe what information was available to those interpreting the index test:</i> <i>If a threshold was used, describe how this was selected (e.g. standard pre-specified threshold, defined a priori within the study, derived from the data):</i>	
2.1 Was the index test conducted and interpreted according to the recommended instructions?	Y/PY/PN/N/NI
2.2 Were the index test results interpreted without knowledge of the reference standard results?	Y/PY/PN/N/NI
2.3 Were the index test results interpreted with the same information as would be available when the test is used in practice?	Y/PY/PN/N/NI
2.4 If an index test threshold was used, was it standard or pre-specified?	Y/PY/PN/N/NI
Risk that the conduct or interpretation of the index test has introduced bias	LOW/HIGH/II
Rationale for judgement	
B. Concerns regarding applicability to the systematic review synthesis question	
Concern that the index test, its conduct, or interpretation does not match the ideal test accuracy trial	LOW/HIGH/II
Rationale for judgement <i>Describe how differences between the setting in which the test was conducted and who conducted and interpreted the test and the ideal test accuracy trial defined in Table 4 have led to this judgement</i>	

Y = yes; PY = probably yes; N = no; PN = probably no; NI = no information; II = insufficient information

DOMAIN 3: TARGET CONDITION

A. Risk of bias	
<i>Define the target condition:</i> <i>Describe how the target condition was assessed – what reference standard(s) was used and how was it conducted and interpreted:</i> <i>What was the time interval between index test and reference standard:</i>	
3.1 Does the reference standard adequately identify those with and without the target condition?	Y/PY/PN/N/NI
3.2 Was the target condition assessed in all participants?	Y/PY/PN/N/NI
3.3 Was the target condition assessed in the same way in all participants?	Y/PY/PN/N/NI
3.4 Did the reference standard avoid incorporating the index test?	Y/PY/PN/N/NI
3.5 Was the reference standard conducted and interpreted according to the recommended instructions?	Y/PY/PN/N/NI
3.6 Were the reference standard results interpreted without knowledge of the index test results?	Y/PY/PN/N/NI
3.7 If a reference standard threshold was used, was it standard or pre-specified?	Y/PY/PN/N/NI
3.8 Was there an appropriate time interval between index test and reference standard?	Y/PY/PN/N/NI
Risk that the assessment of the target condition has introduced bias	LOW/HIGH/II
Rationale for judgement	
B. Concerns regarding applicability to the systematic review synthesis question	
Concern that the target condition as defined by the reference standard does not match the ideal test accuracy trial	LOW/HIGH/II
Rationale for judgement <i>Describe how any differences between how the target condition was defined and the ideal test accuracy trial defined in Table 4 have led to this judgement</i>	

Y = yes; PY = probably yes; N = no; PN = probably no; NI = no information; II = insufficient information

DOMAIN 4: ANALYSIS

A. Risk of bias	
<i>Describe any participants who were enrolled in the study but excluded from the 2x2 table (refer to flow diagram) and reasons why they were excluded (e.g. did not receive index test, did not receive reference standard, uninterpretable index result etc.).</i> <i>How was any missing 2x2 data, including missing reference standard data, accounted for?</i> <i>What do the numbers in the 2x2 table correspond to? That is, what is the unit of analysis (e.g. participants, samples, specimens)?</i> <i>Describe any non-standard statistical analyses used to generate the estimates of sensitivity and specificity being assessed (either conducted by the study authors or by you):</i>	
4.1 Were all participants included in the analysis?	Y/PY/PN/N/NI
4.2 Were missing data handled appropriately?	Y/PY/PN/N/NI
4.3 Does the unit of analysis match the ideal test accuracy trial?	Y/PY/PN/N/NI
4.4 Were the estimates of sensitivity and specificity calculated appropriately?	Y/PY/PN/N/NI
Risk that the analysis has introduced bias	LOW/HIGH/II
Rationale for judgement	

Y = yes; PY = probably yes; N = no; PN = probably no; NI = no information; II = insufficient information

Phase 6: Overall Judgement [for each selected estimate]

An overall judgement across domains should be completed for each estimate specified in Table 5, based on the domain-level judgements. This should be done separately for risk of bias and concerns regarding applicability.

The overall risk-of-bias judgements are based on the domain-level judgements:

- If any domain is judged at high risk of bias, the overall paired estimates are also high risk.
- If all domains are judged low risk, the overall paired estimates are low risk.
- If any domain is judged insufficient information and none are high risk, the overall paired estimates are insufficient information.

Reviewers should also provide a rationale, highlighting the major limitations that informed the overall judgement. A similar approach applies to overall applicability judgements.

A. Risk of bias

Domain	Judgement	Rationale for judgement
1: PARTICIPANTS		
2: INDEX TEST(S)		
3: TARGET CONDITION		
4: ANALYSIS		
OVERALL		

B. Concerns regarding applicability

Domain	Judgement	Rationale for judgement
1: PARTICIPANTS		
2: INDEX TEST(S)		
3: TARGET CONDITION		
OVERALL		